

Ethical Hacking – Personal Study Notes

****These notes are entirely written by me and represent my personal understanding of ethical hacking concepts.****

****I have got another file mentioning all the steps for executing the Tools/Modules**.**

Network Discovery & Scanning

NETDISCOVER

Used to discover live devices inside a local network (LAN).

Command: `netdiscover -i eth0 -r 192.168.1.0/24`

--> Shows IP addresses connected to the same network.

Defense: Network segmentation and monitoring reduce visibility.

NMAP

Network scanning and enumeration tool.

Used to discover open ports, running services, and versions.

`nmap -Pn` --> Skip host discovery.

`nmap -p-` --> Scan all 65,535 TCP ports.

`nmap -sV -sC` --> Service detection + default scripts.

Defense: Firewalls, IDS/IPS, and closing unused ports.

Anonymity & Traffic Routing

PROXYCHAINS

Forces applications to route traffic through proxies like TOR.

`proxychains firefox` --> Launch Firefox through TOR network.

Defense: Proxy detection and traffic analysis.

TOR

Provides anonymity by routing traffic through multiple nodes.

`sudo systemctl start tor` --> Start TOR service.

Note: TOR hides IP, not behavior.

NGROK

Creates a public tunnel to a local service.

Used when attacker and target are on different networks.

`ngrok http 4242` --> Exposes local port 4242 to the internet.

Defense: Block ngrok domains, detect unusual outbound traffic.

Important: Ngrok does NOT exploit systems by itself.

Web Enumeration

GOBUSTER

Brute-forces hidden directories and files.

```
gobuster dir -u http://target -w wordlist.txt
```

Defense: Disable directory listing, use WAF.

FEROXBUSTER

Similar to gobuster, faster and recursive.

```
feroxbuster -u http://target -w wordlist.txt
```

Man In The Middle (MITM)

ARP SPOOFING

ARP has no authentication, which allows spoofing.

Used to intercept traffic inside LAN.

```
arp spoof -i eth0 -t victim router
```

Defense: HTTPS, VPN, Dynamic ARP Inspection.

WIRESHARK

Packet analyzer to inspect network traffic.

GET --> Request data.

POST --> Send data.

Defense: Encryption makes traffic unreadable.

BETTERCAP

Advanced MITM and network attack framework.

arp.spoof on --> Enable MITM.

net.sniff on --> Capture packets.

Defense: Network monitoring, segmentation.

Post Exploitation

REVERSE SHELL

Allows remote command execution if a vulnerability exists.

```
nc -lvnp 4444 --> Start listener.
```

Defense: EDR, outbound firewall rules.

PRIVILEGE ESCALATION

```
find / -perm -u=s 2>/dev/null --> Find SUID binaries.
```

Defense: Patch systems, remove unnecessary SUID files.

Malware Concepts

MSFVENOM

Used to generate payloads.

`msfvenom -p windows/meterpreter/reverse_tcp ...`

Defense: Antivirus, application control.

PERSISTENCE

Backdoors maintain access after reboot.

Defense: Startup monitoring, EDR.

Password Security

HASHING

argon2 --> Strong password hashing.

md5 --> Weak, used only for file integrity.

Defense: Use Argon2, bcrypt, salting.

HASHCAT

Used to crack hashes using wordlists.

Defense: Strong hashing + rate limiting.

OSINT

SUBLIST3R

Finds subdomains of a target.

Subdomains are often less secured.

Defense: Asset inventory and monitoring.

EXIFTOOL

Extracts metadata from images.

Defense: Remove metadata before publishing images.

Legal & Ethical Note

All techniques listed are for educational use only.

Testing must be done only on systems you own or have permission to test.